



Internship report (Tech3)



Developer Low Code Microsoft Power Apps for Ponticelli

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End: 7th of June 2024

Acknowledgement

I would like to thank Franck GUELLAUD, my tutor who helped me and taught me a lot during this internship.

I would also like to thank all the peoples who helped me integrate in the company.

Finally, I would like to thank the Q3SE service for all the documentation and information needed that they have gave me.

Introduction

This internship was a mandatory for the end of my third year at Epitech technology. It was done in Ponticelli in Bassens.

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Part one: Understand the job and tasks.

1. The company

a. A little bit of history

Ponticelli Frères was created in 1921 in Paris by 3 Italian brothers. They gradually became specialised in lifting-rigging activities, mainly in petrol refineries. In 1950 Ponticelli started its industrial piping activities. In the same decade they began to expand their work area by opening regional branches and in the next decades with overseas construction sites. 1970 for Congo, 1973 in Tunisia and so on.

At the turn of the 20th century a strategy of diversifying activities was conducted by adapting internal organisation and reinforced by the acquisition of companies which were experts in their domain, mainly in design but also in piping, lifting, maintenance, etc. In the same time projects multiplied abroad in Africa, Asia and the Middle East.

b. Now

Now in 2024, Ponticelli Frères is an independent and family-owned group providing industrial services mainly to companies in the oil and gas, energy, chemical, pharmaceutical and steelworks sectors.

They generated in 2023 1,234 M€ of turnover. They do mostly work in engineering, construction and maintenance in 5 different markets. The 1st one is Oil and Gas, Petrochemistry and Chemistry. Next there is Nuclear and conventional energies. After that it's Renewable energies. Without forgetting Life sciences Industries. And finally, Infrastructures and Defence. For this work they have more than 6 000 employees all around the world of which more than 3 500 in France. All of that in 80 locations in more than 20 countries.

c. The Great West agency

This internship was performed in the Great West agency which is in Bassens. It manages construction sites and workshops from Bordeaux to Cherbourg. I was in the IT department that was constituted of my tutor Franck GUELLAUD and me. He oversees the IT department of the Great West agency of Ponticelli. I oversaw a project for the Q3SE department.

d. The Q3SE department

They oversee the quality, the security, the safety, and the environment matters. To ensure these matters and avoid problems they have multiple processes. Most of the processes are a form on a paper sheet that are filled in a construction site or in a workshop at the start or the end of a project, an operation, a day. It is made to have feedback and know what the root of a problem is if a problem occurred or to prevent a future problem.

2. The project

a. The start of the development

The project was to develop an application. The development this project started in February 2024 with Microsoft Power Apps which is a low-code solution to create mobile or web applications. We mostly used Microsoft SharePoint to store the application's data. The development started after a period of self-formation using the tutorial available on the Microsoft's page.

b. The organization of the team

My main task was to develop the application and make it work. My tutor supervised the development and used Microsoft Power BI to visualise the data of the application.

I developed in a power apps environment, every time I developed a new feature I shared my version with my tutor. He tested the version and made feedback of the new feature or of a bug that he found.

c. The must

The application must be working in an environment where there was possibly no connection or an instable connection, so for each part I made that it saves in the app and when there is a connection it synchronizes with the SharePoint. It will also use on multiple devices so it must be a responsive design, for all the device to have all the element of each screen.

d. The partition

The application is divided into 8 parts which contain one or more forms. In the main screen we have 8 buttons, one for each part, and they redirect to another screen which contains either a form or another choice of buttons on the different options possible in this part.

3. The tools

a. Microsoft PowerApps

During this internship, I started by using Microsoft PowerApps by discovering and experimenting every new functionality ask by my tutor and the Q3SE service. It's a powerful tool even if it can be limited by being an online and cloud-based solution. Another problem that I encountered is the update of PowerApps and not the documentation of the function and the tutorials that use a function that has been modified.

b. Microsoft SharePoint

For all the data storage, we used SharePoint which is practical because you can use it to create List. And use PowerApps to fill in the lists and after, we can see it online or use a tool like Microsoft Power Bi like my tutor to visualise the data. Or even you can use Microsoft Power Automate to process data.

c. Microsoft Power Automate

I used Power Automate mostly at the end of my internship to make an approval flow. In Microsoft Power Automate, you can create flow that are triggered by a lot of ways, like by example when a new record is created in a Microsoft SharePoint list. And in the flow, you can add lots of action that to modify the newly created element, send email and much more.

d. Microsoft Outlook

In this internship, I used Outlook to communicate by mail, and to manage my agenda. To each new version of the application or the flow, I emailed my tutor with the package of the application or the flow, for him to test it.

Part two: The new project

Hello, since it's the end of my internship and the project was unfortunately not finished, I would like to be hired to be able to finish this project. Seeing that now I have developed the skills in this internship. This new opportunity can be beneficial for both of us, since I already know the project and the company, I will not require to be trained. And this will allow me to improve the quality of what I have done at the start of the project with the newly skill obtained during this internship.

I hope we will work again together, it was a great experience and during this 4 month I learned a lot. Thank you for your time and your work.